Student-Centric Redesign of a Digital Wallet App

CA-3 | Design of Interface Creation with Wireframe Prototyping (CO-4 & CO-5)

Original student-focused UX study inspired by digital-wallet functionality in Nepal; not an exact eSewa clone.

01 — PERSONA

Aarav Sharma • 21 • College Student

Kathmandu, Nepal • Comfortable with smartphones and digital payments.

COMMON STUDENT ACTIVITIES
• Hostel / mess sharing • QR payments • Mobile recharge • Bills • Family requests • Campus expenses

MAIN STUDENT PROBLEMS • Shared expenses are repetitive • Family requests are frequent • Campus payments are scattered • Limited budgets need simple visibility

02 — PROBLEM + UX GOAL

Problem: students complete many small, recurring payments but lack a student-focused organization.

UX GOAL

“Make everyday digital payments simple, fast, understandable and student-friendly.” Built around student life.

03 — USER NEEDS + IA

- Check wallet balance
- Send / receive money
- Scan & pay QR
- Recharge NTC / Ncell
- Pay common bills
- College-related payments
- Review transactions
- Understand success clearly

IA principle: frequent actions first; secondary services stay grouped and discoverable.

INFORMATION ARCHITECTURE + USER FLOW

01 ACCOUNT SETUP
Sign up · OTP · Payment PIN

→

02 WALLET READY
Link bank · View balance · Add money

→

03 STUDENT HUB
Split bills · Request money · Budget

→

04 MAKE A PAYMENT
Send · Review · Authorize · Success

Information Architecture + User Flow

Authentication → Account Setup → Wallet → Transaction → Recovery

AUTHENTICATION	Login • Sign Up • Forgot Password • OTP
ACCOUNT	Payment PIN • Bank Account
WALLET	Home • Send • Receive • Scan & Pay • Bills • Recharge
STUDENT HUB	Student Hub • Split Expenses • Request Money • Campus Pay • Student Budget
TRANSACTION	Review • Payment PIN • Success

RECOVERY • Login → Forgot Password → OTP

HCI Principles Applied

The interface decisions are tied to observable user needs rather than decoration.

1. Visibility of system status	OTP resend timer, linked-bank status, transaction success and current balance are visible when needed.
2. Match with the real world	Terms such as Send Money, Bills, Mobile Recharge, payment PIN and wallet balance match familiar payment tasks.
3. User control & freedom	Back actions and Edit before confirmation let users recover without restarting the task.
4. Consistency & standards	Buttons, fields, status states, spacing and labels reuse the same patterns across the app.
5. Error prevention	Review appears before PIN authorization; fee/source/recipient are shown before commitment.
6. Recognition over recall	Service labels, bank names, categories and student shortcuts are visible instead of hidden behind menus.
7. Flexibility & efficiency	Quick actions and preset recharge amounts reduce repetitive work for frequent student tasks.
8. Aesthetic & minimalist design	Secondary copy is quieter; primary actions are visually dominant; unrelated detail is kept out of the path.
9. Help users recover	Forgot Password, OTP guidance and security notes provide recovery/help without interrupting the main task.
10. Accessibility	Readable text sizes, adequate contrast, explicit labels and clear states avoid relying on colour alone.

Final QA: clear hierarchy • predictable navigation • visible status • error prevention • user control • recognition over recall

PaySathi — Design System & UX Principles

Second-pass refinement: restrained visual language, realistic financial UX, reusable components and explainable interaction decisions.

COLOUR TOKENS

Primary Cobalt

Deep Navy

Ice Blue

Neutral Slate

Navy Text

Error Red

Positive Teal

TYPE HIERARCHY

Heading / 24–30 px

Section / 15–16 px

Body / 11–13 px

Caption / 8–10 px

COMPONENTS

- Button
- Input
- Quick Action
- Transaction Row
- Wallet Card
- Status
- Bottom Navigation
- Add Money Sheet

UX PRINCIPLES APPLIED

1. Visual Hierarchy	Important financial information receives stronger visual emphasis.
2. Consistency	Buttons, inputs and navigation maintain consistent patterns.
3. Recognition	Icons and labels help users identify actions without memorization.
4. Error Prevention	Users review transaction details before authorization.
5. Feedback	Successful transactions provide immediate confirmation.
6. Progressive Disclosure	Only necessary information is shown at each stage.
7. Accessibility	Readable typography, contrast and touch-friendly controls improve usability.

DESIGN INTENT

Simple, secure, student-friendly digital payments.

Navy establishes trust; cobalt identifies primary actions and selected states; teal communicates positive outcomes; amber is reserved for due dates and important alerts.

FLOW REFINEMENT

LOGIN → HOME → SEND MONEY → REVIEW & CONFIRM → PAYMENT PIN → PAYMENT SUCCESS

Recovery: LOGIN → FORGOT PASSWORD → OTP VERIFICATION → HOME

Other services: HOME → SCAN & PAY / BILLS / MOBILE RECHARGE

UX Annotations — Design Rationale

Each annotation explains WHY the interface uses a specific pattern.

A1 — Persona relevance	Student needs drive information architecture.
A2 — Login choice	Mobile / Email provides familiar entry points.
A3 — Quick actions	Frequent tasks are surfaced early.
A4 — Wallet balance	Financial context is prominent.
A5 — Linked bank	Funding context is visible before payment.
A6 — Error prevention	Review precedes authorization.
A7 — User control	Edit is available before confirmation.
A8 — PIN security	Authorization is isolated.
A9 — QR discoverability	Scan & Pay is a primary action.
A10 — Bill grouping	Categories reduce search effort.
A11 — Recharge shortcuts	Preset amounts reduce typing.
A12 — Success feedback	Receipt details make outcome explicit.
A13 — Accessibility	Readable text and usable controls support broader use.
A14 — Consistency	Repeated actions share the same visual language.
A15 — Student-Centric Feature Grouping	Student Hub groups frequent student-specific financial tasks into one discoverable location.
A16 — Shared Expense Support	Split Expenses supports common hostel and mess payment scenarios.
A17 — Spending Awareness	Student Budget provides a lightweight spending overview without complex analytics.

9:41

PaySathi

PAYMENTS MADE FOR CAMPUS LIFE

Your campus life. Money made easy.

Pay friends, split the mess bill, and recharge in seconds.



Mobile | Email

Mobile number

+977 98XXXXXXXX

Get Started →

Forgot Password?

Create account

Protected by secure PIN + verified mobile

9:41

Create your account

Join PaySathi in under a minute

01 Your details

STEP 1 OF 4

Mobile | Email

Full name

Aarav Sharma

Mobile number

+977 98XXXXXXXX

Email address (optional)

name@example.com

Create password

••••••••••

PASSWORD STRENGTH

Good

I agree to the Terms of Service & Privacy Policy

Continue →

Already have an account? Log in

Your account, your control

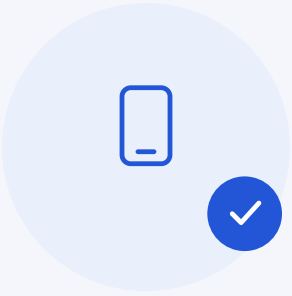
Verify your number, set a payment PIN, then connect a bank account.

NEXT • VERIFY YOUR MOBILE

9:41

Verify your number

A quick security check to protect your wallet



Code sent to +977 98••••••XX

Enter the six-digit verification code

4 8

Resend code in 00:30

Resend OTP

Verify & continue →

Never share your code. PaySathi uses it only to verify your identity.

1 2 3

4 5 6

7 8 9

0

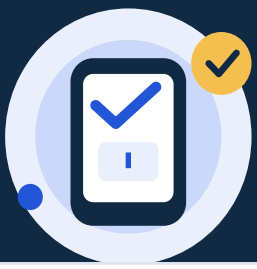
9:41

PaySathi

ACCOUNT RECOVERY

Forgot your password?

Let’s get you back into your wallet safely.



Recover access

We’ll send a secure verification code to your registered mobile number.

Registered mobile number

+977 98XXXXXXXX

Send verification code →

THREE QUICK STEPS

1 Verify — 2 Confirm — 3 Reset

Your account details stay private throughout recovery.

Remembered your password? Log in

No code? Check your number and try again.

9:41

Create your payment PIN

A private four-digit code authorizes payments



SECURITY SETUP · 1 OF 2

4-digit PIN

Confirm PIN

Create secure PIN →

Only you should know this code

Avoid birthdays. Never share your payment PIN.

Your PIN is used only to approve payments.

9:41

Link your bank

Choose your bank to fund your PaySathi wallet

SUPPORTED NEPALI BANKS

Search banks by name

NIC — **NIC Asia Bank** — Popular choice ✓

N — **Nabil Bank** — Fast wallet linking

G — **Global IME Bank** — Secure account connect

H — **Himalayan Bank** — Trusted banking

ACCOUNT DETAILS

Account number

•••• 4821

Account holder name

Aarav Sharma

Your bank stays in control

Account details stay masked in PaySathi.

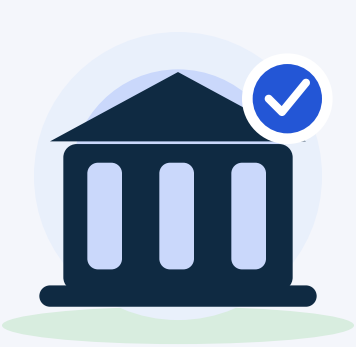
Link bank account →

Prototype only · no real banking integration

YOUR CONTROL, ALWAYS

Masked details | Encrypted | Unlink anytime

9:41



ALL SET

Your bank is connected

Your PaySathi wallet is ready for campus life.

NIC Asia Bank
Savings · •••• 4821

✓ Connected

WHAT WOULD YOU LIKE TO DO?

Add money
Top up wallet

Pay bills
Utilities & fees

Split costs
With friends

Go to my wallet →

Payments, with peace of mind

Review every transfer and approve it with your secure PIN.

You can manage linked accounts from Profile.

No setup fee · Manage anytime · Pay in NPR

9:41

Hey, Aarav

Your money, in a better place.

AS

TOTAL WALLET BALANCE

Rs. 4,250

Available to spend

+ Add money

NIC Asia · •••• 4821

Quick actions

- Send money
- Receive
- Scan & Pay
- Pay bills

MADE FOR STUDENT LIFE

Split bills. Keep it easy.

Hostel, mess, trips & more

Upcoming payments

Mess payment	Hostel · Monthly	Due tomorrow
Internet	WorldLink · Shared	In 3 days

Recent activity

R

Sent to Rahul

Today · 10:42 AM

– Rs. 500

Hostel electricity split

Home | Pay | Campus | Budget

9:41

Send money

Fast transfers, with a clear final check

SEND TO

Mobile number or contact

+977 98XXXXXXXX

RECENT PEOPLE

R Rahul

A Aaditya

K Karan

M Mom

\+ Other

AMOUNT TO SEND

Rs. 500

No hidden surprises

Rs. 100 Rs. 500 Rs. 1,000

What's this for?

Hostel electricity split

PaySathi Wallet

Available · Rs. 4,250

You're sending Rs. 500

To Rahul Sharma · Fee Rs. 0

You'll review the recipient and amount before entering your PIN.

Review transfer →

Secure transfers · You stay in control

TRANSFER JOURNEY

1 Details → 2 Review → 3 Approve

9:41

Review transfer

Take one last look before you send

SENDING TO

R

Rahul Sharma

Rs. 500

FEE Rs. 0

PAYMENT DETAILS

Mobile	98XXXXXXX
Purpose	Hostel electricity
Paying from	PaySathi Wallet
Transaction fee	Rs. 0

Total you'll pay **Rs. 500**

Protected payment

You'll authorize the transfer with your private PIN.

Confirm & continue →

← Edit transfer details

Your money won't move until you approve it.

Review ● → Authorize → Receipt

9:41

Authorize payment

Enter your PIN to finish this transfer

PAYMENT TO RAHUL SHARMA

Rs. 500

Hostel electricity split

ENTER YOUR 4-DIGIT PIN

● ● ● ●

Forgot PIN?

Use the secure keypad below

1	2	3
4	5	6
7	8	9
	0	⌫

Confirm payment →

PaySathi never stores or displays your real PIN in this prototype.

Encrypted journey · You're in control

STEP 2 OF 3

Review ✓ · Authorize ● · Receipt

9:41

PaySathi / PAYMENT COMPLETE



Payment successful!

Rahul received your money.

Rs. 500

PAYMENT RECEIPT ✓ SUCCESS

Paid to	Rahul Sharma
Date	13 Oct 2026
Reference ID	PS-84•••219
From	PaySathi Wallet

↓ Receipt ↗ Share Done →

One less thing to worry about.

Your transaction is saved to Activity.

Keep this receipt for your records.

NEXT UP

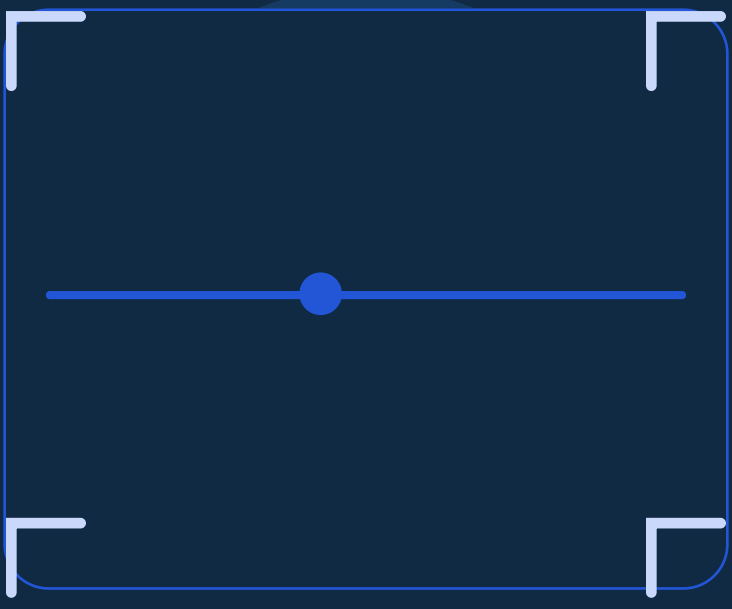
Split another campus expense Open Student Hub →

9:41

Scan & Pay

Pay a merchant or friend using their QR

ALIGN QR WITHIN FRAME



Camera preview shown for the HCI prototype

Flashlight

Upload QR

YOUR PERSONAL QR

Let friends scan to pay you

Receive without sharing your number.

Show my QR →

RECENT MERCHANT

Campus Café

Last paid Rs. 180 · QR payment

Scan again →

Home

Pay

Campus

Budget

Pay bills

Everything due, in one easy place

Search electricity, internet, fees...

QUICK ACCESS

- Electricity
- Internet
- College fees
- Mobile

Utilities & home

- **Electricity** — NEA · Household bill
- **Water** — KUKL · Utility bill
- **Internet** — WorldLink · Shared plan

Campus & communication

- **College fees** — Semester / exam payments

NEEDS YOUR ATTENTION

WorldLink Internet
Due tomorrow
Rs. 1,250
Pay now →

Stay ahead of due dates
Save your regular bills and avoid last-minute stress.

9:41

Mobile recharge

Top up NTC or Ncell in a few taps

NTC | Ncell

MOBILE NUMBER

98XXXXXXXX

RECENT NUMBERS

- 98•••412 NTC
- 98•••830 Ncell
- Mom NTC

CHOOSE AMOUNT

- Rs. 100
- Rs. 200 (POPULAR)
- Rs. 500
- Rs. 1,000

Bonus data pack

1.5 GB data · 7 days

Included

PaySathi Wallet

Balance Rs. 4,250

Fee Rs. 0

We'll show the number and amount before final confirmation.

Recharge Rs. 200 →

Safe payments · No balance surprises

Home | Pay | Campus | Budget

9:41

Student Hub

Campus money, minus the awkward bits.



✦ BUILT FOR STUDENT LIFE

Spend together. Stay in control.

3 of 4 friends settled

Your student toolkit

Simple tools for shared campus life

Split expenses
Hostel & mess bills

Request money
No awkward reminders

Campus pay
Fees & cafeteria

Student budget
See where it goes

Coming up

VIEW ALL →

Mess contribution
Due tomorrow
Rs. 1,800

ONE THING TO WRAP UP

Aaditya still owes Rs. 300
Send a kind reminder

Remind →

Home | Pay | Campus | Budget

Split an expense

Share a cost fairly, without the awkward math

HOSTEL ELECTRICITY

September bill

TOTAL BILL

Rs. 1,200

4 PEOPLE

A R A K

Your group

Rs. 300 per person

A	Aarav — YOU PAID	Rs. 300 ✓
R	Rahul — PAID	Rs. 300 ✓
A	Aaditya — PENDING	Rs. 300 ●
K	Karan — PAID	Rs. 300 ✓

SETTLEMENT PROGRESS — 3 of 4 paid

Send reminder to Aaditya →

+ Create a new split

Amounts are shown equally for this example.

Student budget

Your monthly plan, at a glance

October 2026 — This month

MONTHLY BUDGET

Rs. 8,000

Planned for October

SPENT Rs. 4,250 LEFT Rs. 3,750

Where your money goes

Share of spending

F	Food & mess	Rs. 2,100
T	Transport	Rs. 850
M	Mobile & data	Rs. 300
+	Other	Rs. 1,000

SMART TIP

You've got Rs. 3,750 left.

Keep some room for the unexpected.

LAST TRANSACTION

Campus cafeteria

Today · Food & mess

– Rs. 240